Assignment 1: Branch Predictor Implementation

Course 6385

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1 What was implemented

The predictor is the class `pm_predictor` in `cbp2-infrastructure-v2/src/my_predictor.h`. It is the simplified hybrid from Appendix B of the assignment: a bimodal table and a two-way global table. The assignment allows a two-way global predictor in place of the Pentium M's four-way table.

A conditional branch is predicted as follows.

- The bimodal table has 16 two-bit saturating counters. The index is `PC[5:2]`. The prediction is the counter's high bit.
- A 4-bit pattern history register records the outcomes of conditional branches. This is the global history register in Appendix B. It is shifted after the branch resolves, and the new taken or not-taken bit is inserted.
- Those four history bits are XORed with `PC[5:2]`. Bits `[3:2]` of the result select one of four sets. Bits `[1:0]` are the tag.
- Each set has two ways. A way hits when it is valid and the tag matches. On a hit the global counter's high bit is the prediction. On a miss the bimodal prediction is used.
- The global entry is updated either way. A miss fills an invalid way, or replaces the least-recently-used way if both are valid. The chosen way becomes most recently used.

Counters saturate at 0 and 3. The initial state matches the Appendix B tables: bimodal counter i starts at $i \bmod 4$, and the global ways start with the valid bits clear and the counter and LRU values shown in that example.

The pattern history register is the simple stand-in for the Pentium M path information register. Both are XORed with the branch address to index the global table. The path register stores address and target bits and is updated only on taken and indirect branches. This design stores one outcome bit per conditional branch.

2 What was left out

Loop prediction, the branch target buffer, and the path information register are not implemented. Unconditional branches are predicted taken and do not update the tables. Target addresses are not predicted; the simulator still reports direction mispredictions per kilo-instruction, which is the metric for this assignment.

3 Check against Appendix B

Appendix B gives 22 conditional branches (`b1` at `0x44`, `b2` at `0x6C`, `b3` at `0x90`) and the expected state of both tables. Those tables are stored as `hybrid.csv`, `local.csv`, and `global.csv`. The program `appendix_b_check` feeds each program counter and outcome to `pm_predictor` and compares the history register, the XOR index, the 16 bimodal counters, and every global valid bit, tag, counter, and LRU bit.

The run on the current predictor reports 22 of 22 rows matching.

4 Trace results

The predictor was built with `make` in `cbp2-infrastructure-v2/src` and run on the 20 CBP2 traces. Each trace is 100 million instructions. MPKI is $1000 \times (\textit{direction mispredictions})/10^8$, counted only for conditional branches. The always-taken column is the same traces run with the original stub, which predicted every conditional branch taken.

Trace	Always taken	Hybrid
164.gzip	70.025	49.708
175.vpr	85.312	40.103
176.gcc	86.071	61.200
181.mcf	102.497	41.210
186.crafty	48.651	40.400
197.parser	78.421	38.632
201.compress	76.983	26.885
202.jess	92.343	47.089
205.raytrace	109.506	11.827
209.db	91.451	32.173
213.javac	109.644	13.131
222.mpegaudio	110.339	12.785
227.mtrt	112.022	12.275
228.jack	77.958	41.810
252.eon	24.226	22.745
253.perlbmk	66.833	57.871
254.gap	75.950	44.542
255.vortex	70.524	38.671
256.bzip2	178.257	77.119
300.twolf	59.221	45.328
Average	86.311	37.775

Table 1: Direction mispredictions per kilo-instruction.

The hybrid predictor is lower than always-taken on every trace. The average falls from 86.311 to 37.775 MPKI. The largest drops are on raytrace, javac, mpegaudio, and mtrt, which are heavily biased and fit even a small bimodal table. gcc, perlbmk, and bzip2 stay high. The global table has only four sets and two tags, so unrelated branches alias, and a hit can override a correct bimodal prediction with a counter trained by a different branch.

5 Conclusion

The submitted predictor is the Appendix B hybrid: a 16-entry bimodal table, a 4-bit pattern history register, and a 2-way tagged global table. A global hit overrides the bimodal prediction. The implementation matches the worked example cell for cell, and on the CBP2 traces it cuts the always-taken miss rate from 86.311 to 37.775 MPKI.